

Final Project - Reliability in ML

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1 Abstract

This project extends the work on Conformalized Unconditional Quantile Regression [Alaa et al. \[2023\]](#) to enhance its robustness against data distribution shifts. While previous research has suggested that CUQR demonstrates resilience to such shifts, we aim to further improve this aspect of the algorithm. Our approach involves refining the base CUQR algorithm through three main strategies: (1) implementing dynamic group insertion to adapt to evolving data distributions, (2) developing methods for adaptive parameter adjustment, and (3) introducing dynamic online updates for real-time adaptation. We hypothesize that these modifications will significantly increase CUQR's adaptability to varying data distributions. Our methodology combines the implementation of these adaptive mechanisms with empirical evaluation across diverse datasets to assess the effectiveness of our proposed enhancements. This research contributes to the ongoing efforts to develop reliable machine learning techniques that maintain performance under real-world conditions where data distributions may fluctuate unpredictably.

2 Background and problem setup

The field of machine learning has made significant strides in predictive modeling, yet the challenge of reliable uncertainty quantification remains a critical area of research. This challenge is particularly pronounced in scenarios involving heterogeneous data distributions, where the relationship between features and outcomes can vary significantly across the covariate space. Traditional predictive inference methods often struggle to adapt to these varying levels of uncertainty, resulting in predictions that may be overly conservative in some regions of the feature space and unreliable in others. This lack of adaptability can lead to several issues:

1. Inefficient resource allocation in decision-making processes.
2. Reduced trust in model predictions due to inconsistent reliability.

3. Potential overlooking of critical cases where uncertainty is underestimated.

Furthermore, many existing methods provide only marginal coverage guarantees, which, while valuable, do not offer insights into the model’s performance in specific regions of the covariate space. This limitation can be particularly problematic in high-stakes applications where understanding localized model confidence is crucial. Our research aims to address these challenges by building upon recent advancements in adaptive conformal prediction. Specifically, we focus on enhancing the robustness of these methods to significant data distribution shifts, a scenario commonly encountered in real-world applications where training data may not fully represent future operational conditions.

3 Conformalized Unconditional Quantile Regression

Conformalized Unconditional Quantile Regression (CUQR), introduced by [Alaa et al. \[2023\]](#), represents a significant advancement in addressing challenges in predictive interval estimation. CUQR builds upon the framework of conformal prediction, offering distribution-free marginal validity guarantees, and extends it to provide more granular, instance-specific predictive intervals. The key innovation of CUQR lies in its approach to partitioning the covariate space into subgroups. This partitioning allows the method to:

1. **Adapt to Heteroscedasticity:** By dividing the covariate space, CUQR can capture and respond to varying levels of uncertainty across different regions.
2. **Enhance Transparency:** CUQR not only provides predictive intervals but also reports the specific subgroup to which a prediction belongs.
3. **Balance Granularity and Stability:** The method introduces a trade-off between the granularity of subgroups and the stability of coverage guarantees.
4. **Offer Robust Guarantees:** CUQR aims to deliver finite-sample coverage guarantees within each subgroup.

CUQR operates by first partitioning the covariate space into subgroups, typically using clustering algorithms. It then applies a two-step process:

1. **Unconditional Quantile Regression (UQR):** This step involves fitting a machine learning model to predict the re-centered influence function (RIF) of the quantile functional.
2. **Conformal Calibration:** CUQR selects the tightest predictive band that achieves the desired coverage within each subgroup using a held-out calibration set.

Despite these advancements, the robustness of CUQR to significant data distribution shifts remains an area for potential improvement.

4 Extensions

Building upon the foundational work of Conformalized Unconditional Quantile Regression (CUQR), we have implemented several extensions to enhance its robustness against significant data distribution shifts. Our approach focused on two primary areas of improvement:

1. **Dynamic Group Insertion** We implemented a flexible mechanism for dynamically expanding the model’s group structure in response to evolving data distributions. This enhancement includes: A simple group creation method using k-means clustering on newly observed data, providing a basic partitioning of the expanded feature space while not compromising the previous groups fitted on the old data.
2. **Adaptive Parameter Adjustment** We developed sophisticated methods for dynamically updating CUQR’s key parameters to improve its adaptability to varying data distributions. This extension includes: We implemented functions that update g_θ, q_p , and f_e based on new incoming data, allowing the model to continuously adapt to changing data distributions.
3. **Dynamic Online Updates** We implemented an online update mechanism triggered when the average coverage of any subgroup falls below a predefined threshold (0.85). Upon activation, the algorithm creates a set number of new subgroups (e.g., three) using k-means clustering on recent data and updates the model’s internal functions (g_θ, Q_P, f_E). This enables the model to adjust and maintain coverage as data distributions evolve.

These extensions have significantly improved CUQR’s performance and adaptability in the face of distribution shifts, as demonstrated by our experimental results.

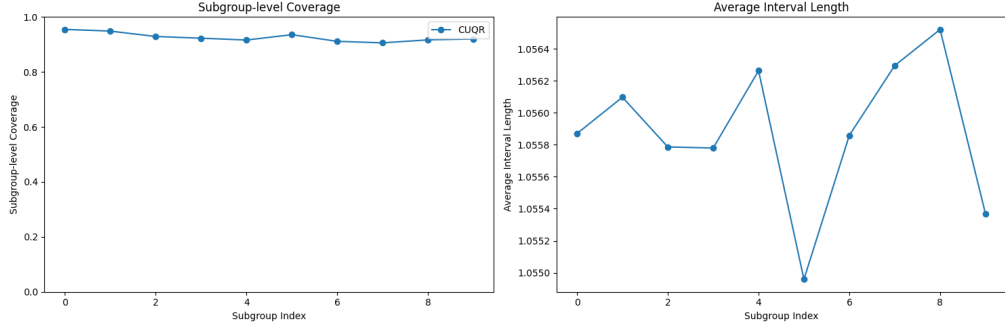
5 Results

In all of our experiments we used $\alpha = 0.1$

5.1 Reproduce results on MEPS-19

We first implemented the CUQR algorithm described in [Alaa et al., 2023]. We used the Medical Expenditure Panel Survey (MEPS) data from 2019 [Agency for Healthcare Research and Quality \[2019\]](#), which provides detailed information about healthcare utilization and expenditures in the United States.

Important note: We weren’t sure if we need to scale the data so that it would work well for the algorithm, we scaled it and then we didn’t know exactly how to scale back the interval lengths so we got minor changes in the interval avg lengths. As we can see the algorithm keeps close to 90% coverage.



5.2 Experiment on Synthetic Dataset

We evaluate our method by creating a synthetic data of 3000 samples sampled from a normal gaussian distribution ($\mu = 0, \sigma = 1$). We measured the results of standard CUQR on this dataset

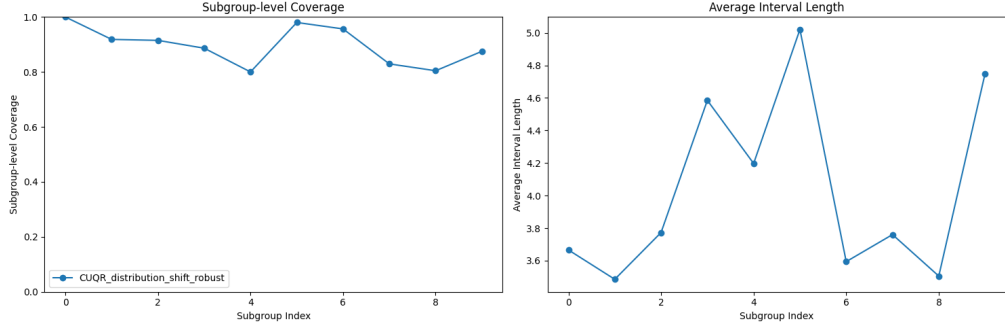


Figure 1: CUQR coverage and interval lengths on the original synthetic data

We conducted two distinct experiments to assess the robustness of our model to distributional shifts in the input features. In both experiments, we systematically altered 50% of the feature set, while leaving the remaining features unchanged. The experiments were designed as follows:

1. **Mean Shift Experiment:** In this experiment, we induced a shift in the average of the selected features. Specifically, we applied an additive perturbation of 0.5 to the mean of each selected feature, effectively translating their distributions. This can be expressed mathematically as:

$$\tilde{x}_i = x_i + 0.5 \cdot 1_{i \in S} \quad (1)$$

where $\tilde{x}_i$ is the perturbed feature, x_i is the original feature, and S is the set of randomly selected features.

2. **Variance Scaling Experiment:** For this experiment, we focused on amplifying the magnitude of the selected features. We multiplied each selected feature by a factor of 2, thereby increasing both their spread and their absolute values. This transformation can be represented as:

$$\tilde{x}_i = \begin{cases} 2x_i, & \text{if } i \in S \\ x_i, & \text{if } i \notin S \end{cases} \quad (2)$$

where $\tilde{x}_i$ is the transformed feature value, x_i is the original feature value, and S is the set of randomly selected features.

These controlled perturbations allowed us to simulate realistic scenarios of data drift and evaluate the model’s performance under such conditions. By selectively altering different statistical properties of the input features, we aimed to gain insights into the model’s sensitivity to various types of distributional changes.

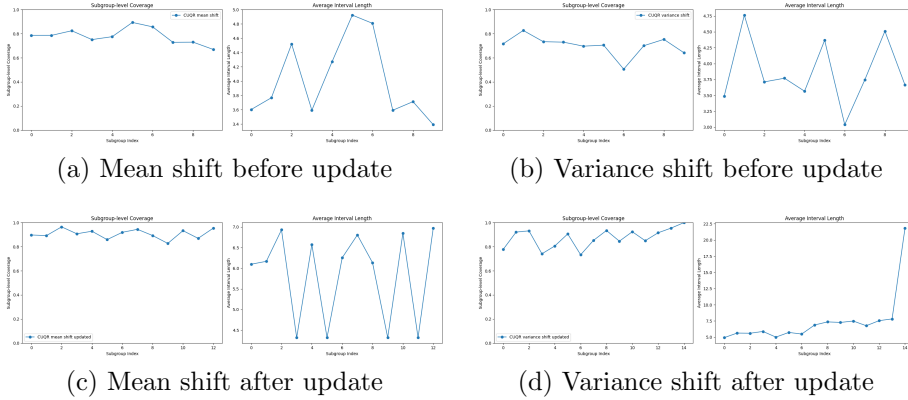


Figure 2: CUQR coverage and interval lengths on synthetic data before and after updates. (a) Shifted mean before update, (b) Scaled variance before update, (c) Shifted mean after update, (d) Scaled variance after update.

We update the model functions $g(\theta)$, Q_P , f_E and then we calculated the conformity scores and tightest bands.

Shift Type	Coverage		Average Length	
	Before	After	Before	After
Mean	0.7804	0.9071	4.0183	5.8498
Variance	0.6707	0.8732	3.8340	7.4207

Table 1: Comparison of coverage and average length before and after updates for mean and variance shifts.

5.3 Experiment on Streaming Synthetic Dataset

In this experiment, we aim to simulate a real-world scenario where the algorithm processes data sequentially, handling each incoming instance one at a time. The primary objective is for the algorithm to detect significant drops in coverage and, upon doing so, initiate updates to halt the decline and ideally restore coverage to previous levels.

For this section, we focus exclusively on data that has undergone a mean shift.

Detailed Experiment Description: We begin by generating synthetic data from a known distribution and evaluating the model’s coverage on an unseen set of data sampled from the same distribution. Although we will not present these plots here, they mirror the results from the previous section. Subsequently, we generate an additional dataset from the same distribution, apply a mean shift to it, and pass the data to the algorithm sequentially, one instance at a time. As the algorithm processes this stream, it continually updates the average coverage for various subgroups.

When the average coverage of any subgroup falls below a predefined threshold (set to 0.85 in this experiment), the algorithm triggers an update. This update introduces three new subgroups (this value can be adjusted as a hyperparameter) and updates the model’s internal functions, following the same procedure as in the previous experiment. We then track the model’s coverage, which, as expected, shows an upward trend after the update.

To provide further insights into the model’s behavior, we will display the average coverage of specific subgroups independently, highlighting the internal dynamics of the algorithm.

Figure 3 displays the average coverage of the model on subgroup 1 (which is an original subgroup of the model). The red dot highlights the point in time where the update was done. And figure 4 displays the average coverage of subgroup 8 (which was added after the update):

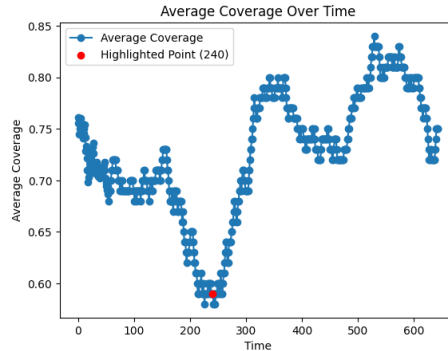


Figure 3: Average Coverage Over Time - Subgroup 1

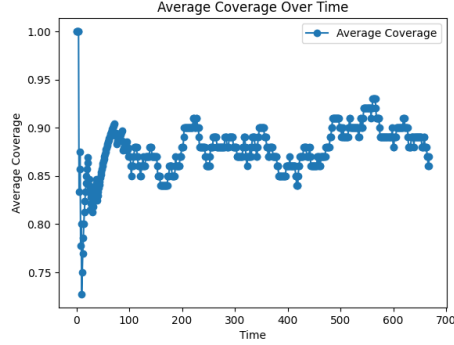


Figure 4: Average Coverage Over Time - Subgroup 8

6 Conclusions

In this work, we extended the Conformalized Unconditional Quantile Regression (CUQR) framework to improve its robustness against significant data distribution shifts. Through dynamic group insertion, adaptive parameter adjustments, and online updates, we have demonstrated the algorithm’s ability to maintain reliable coverage in the presence of covariate shifts. Our experiments showcased how these enhancements allow CUQR to detect performance degradation in subgroups and effectively update its internal models to restore and stabilize coverage.

Specifically, we observed that the algorithm could successfully identify when coverage dropped below a predefined threshold and, after performing updates, regained its desired performance. This demonstrates the potential of our approach for use in real-world applications where data distributions evolve over time, making it crucial to adaptively maintain uncertainty quantification.

These results contribute to the ongoing development of machine learning techniques that ensure reliable performance under dynamic, non-stationary data conditions. Our enhancements to CUQR provide a step forward in building more resilient predictive models capable of handling real-world complexities.

7 Future Work

While our current extensions have shown promising results, there are several avenues for future research that could further enhance the capabilities of CUQR:

7.1 Advanced Hyperparameter Tuning

1. **Adaptive Cross-Validation:** Implementing a novel cross-validation strategy that weights different subgroups based on their susceptibility to distribution shifts. This approach aims to optimize hyperparameters for both overall performance and robustness to potential shifts.

2. Meta-Learning for Hyperparameter Initialization: Utilizing meta-learning techniques to initialize hyperparameters based on dataset characteristics, potentially improving the algorithm’s ability to quickly adapt to new distributions.

7.2 Dynamic Group Refinement

1. Adaptive Group Splitting: Implementing a mechanism to dynamically split groups when significant heterogeneity is detected within a subgroup, potentially indicating the emergence of a new distribution.
2. Group Merging for Stability: Developing criteria for merging similar groups to maintain statistical power when the number of samples in individual groups becomes too small due to distribution shifts.

7.3 Incorporating Drift Detection Mechanisms

1. Integration of Drift Detection Algorithms: Explore the use of more sophisticated drift detection algorithms, such as ADWIN or Page-Hinkley, to automatically trigger model updates when a shift is detected. These mechanisms could complement the current approach of monitoring average coverage and provide a more fine-grained response to distributional changes.
2. Multivariate Drift Detection: Develop methods for detecting multivariate drifts across different subgroups simultaneously, allowing for more nuanced responses to complex shifts that may affect certain features or subgroups disproportionately.

These potential extensions could further improve CUQR’s adaptability and performance in scenarios with complex and evolving data distributions. Future research in these areas could lead to even more robust and flexible predictive models.

References

- Agency for Healthcare Research and Quality. Medical Expenditure Panel Survey 2019. <https://www.meps.ahrq.gov/mepsweb/>, 2019. 3
- Ahmed M. Alaa, Zeshan Hussain, and David Sontag. Conformalized unconditional quantile regression. *arXiv preprint arXiv:2304.01426*, 2023. 1, 2, 3